

Count Piece Picks into Bags SOP

One episode fills four bags. It runs from a bulk bin of loose pieces, a bare count board, four empty bags standing open in the bag rack, a full label strip, a blank tally card and an empty done rack, to four bags each holding exactly five pieces, each one sealed, labelled with its own number and standing in the done rack slot with the same number, with the tally card carrying one stroke in every row and every tool back on its spot.

One bag is finished before the next one is started. The order for each bag never changes: **count five pieces onto the board, move them into the bag, seal the bag, label the bag, verify the tally card and mark it, stand the bag in the done rack.** Nothing goes into a bag until five pieces are lying on the board and can be counted, and no bag is sealed until the board is bare.

One arm does all of it. The same gripper takes every piece, lays it out, moves it into the bag, seals the bag, presses the label on, draws the stroke and racks the bag. Nothing in this task has to be held, because nothing moves on its own: the bin, the board, the bag rack, the seal pad, the tally card and the done rack all grip the table, and each bag stands open on its own in its ring. Only one arm works in this cell. Do not bring a second arm into the task.

The count is the whole job. Pieces dropped straight into a bag cannot be counted back on camera, so every piece is laid out in the open first: one piece per pick, one piece per square, five squares, and only then does anything go into the bag. The board is the count. A bag is never counted by eye, by weight or by a handful.

Every spot sits inside the arm's comfortable reach, so the arm never has to stretch or fold in on itself. Only one object is held at a time, no grasp is changed while an object is off the table, and nothing is ever carried over an open bag, over a finished bag or over the tally card. Each piece travels from the bin to the board and then round the front of the bag rack to its own bag.

The counts are fixed: four bags, five pieces a bag, twenty pieces in total, one seal, one label and one stroke per bag. There is no sixth piece, no second label and no second stroke. A short count is not topped up after the bag is sealed and an over count is not tipped back out: the bag is left as it is, tagged, and the episode ends.

Each tool goes back to its own spot before the next step starts. A piece that lands off the board or off the table is left exactly where it falls, and one fresh piece is taken from the bin to fill its square. Nothing is ever picked up off the floor on camera and no piece is swept up on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the bulk bin, the count board with all five squares, the bag rack with all four bags, the label strip, the seal pad, the tally card, the marker rest and the done rack.
3. The count board stays in frame the whole time, close enough that a reviewer can count the pieces lying on it and read which square each one is in.
4. All five square numbers and all four bag numbers face the camera, and the tally card's four rows can be read from the camera without the arm standing over them.
5. The mouth of the bag being filled stays in frame for the whole of the move, so a reviewer can see each piece go in.
6. The light falls across the board and the seal pad without glare, so a piece on a square, the closed seal strip and the ink of a fresh stroke can all be seen.
7. The arm is at home with its gripper open.
8. The tabletop is dry and clear of anything but the items in the Materials checklist.
9. The arm reaches every spot on the table, the bin, all five squares, all four bag mouths, the label strip, the seal pad, the tally card, the marker rest and all four done rack slots, without extending to a joint limit or folding in on itself. Bring the spots forward or closer together until this is true.
10. The arm comes down straight from above onto the bin, onto every square, onto every bag mouth, onto the seal pad and onto the marker rest, so a piece is lifted and lowered without being tilted or swung in.
11. The arm reaches each bag mouth from the front of the rack without passing over the mouth of another bag. Move the rack forward until this is true.
12. The paths from the bin to the board, from the board round the front of the rack to each bag, from the rack forward to the seal pad, and from the seal pad to each done rack slot are unobstructed. Carry one piece and one empty bag along each path by hand during setup: they must clear the bin rim, the board edge, the rack rings and the tally card.

Materials checklist

1. One bulk bin stands at the back of the table, straddling the midline, holding loose pieces and nothing else.
2. The bin grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or put it in a holder. Nothing pins the bin during the episode.

3. The bin holds well more than twenty pieces, spread across its floor rather than heaped in one corner, and shallow enough that the gripper reaches a piece without passing below the bin rim.
4. Every piece in the bin is the same part in the same size and colour, so any piece fits any bag and no piece has to be chosen over another.
5. Each piece is big enough for the gripper to close on squarely and to be seen on camera as one piece, and small enough that five of them sit in a bag well below its seal strip. Pieces that stick, nest or hook together are replaced: two pieces that come up as one cannot be counted.
6. One count board lies flat left of the midline in the middle row, with five squares in one line, numbered 1 to 5 left to right. The numbers face the camera.
7. The board grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or put it in a holder. Nothing pins the board during the episode.
8. Each square is wide enough to hold one piece with clear board showing round it, and shallow enough that a piece lifts straight up out of it. A square that holds two pieces side by side is too wide and the board is replaced.
9. The board is bare at the start of the episode and bare again at the end.
10. One bag rack stands right of the midline in the middle row, holding four bags in one line, numbered 1 to 4 left to right with the numbers facing the camera.
11. Four poly bags, and no more, one hanging in each numbered ring of the rack. Every bag is the same size and every bag carries a press to close seal strip across its mouth.
12. Each bag hangs open on its own in its ring, its mouth held wide enough for the gripper to lower a piece inside without touching the bag walls, and its seal strip standing open and clear above the ring.
13. Every bag is empty, dry and clear inside at the start of the episode, with nothing already in it and no tear or hole in it.
14. Each bag lifts straight up out of its ring by hand during setup without the ring holding it back and without the mouth closing on the way out. A rack that grips the bags is replaced.
15. A clear gap of at least a gripper's width sits between one bag and the next, so the arm comes down on one mouth without touching the bag beside it.
16. One seal pad lies flat right of the midline in the front row, straight forward of the bag rack, bare at the start of the episode and bare again at the end. It is firm and flat, so a bag laid on it is pressed against the pad and not against bare table.
17. One label strip lies flat at the right end of the middle row, clear of the bag rack, carrying four labels on its backing, numbered 1 to 4 in the order they are peeled. The numbers face the camera.

18. Each label peels off its backing under a straight lift from one edge, and sticks to a poly bag under one press. Peel a spare label by hand during setup: a label that tears, that lifts its backing with it, or that will not stick to poly is replaced.
19. One tally card lies flat left of the midline in the front row, carrying four rows numbered 1 to 4 top to bottom. Each row prints its bag number and the count 5, and ends in one empty box. The card is blank in every box at the start of the episode.
20. The card grips the table or sits in a holder. Push it by hand during setup: if it slides, turns or lifts, weight it. Nothing pins the card during the episode.
21. One dry marker lies flat on its rest at the left end of the front row, cap off, tip toward the front of the table, clear of the tally card. It marks poly and card under one stroke and does not need to be shaken or primed.
22. One done rack stands at the right end of the front row with four slots in one line, numbered 1 to 4 left to right with the numbers facing the camera. It is empty at the start of the episode and holds four bags at the end.
23. Each done rack slot holds one sealed bag standing up on its own, with its labelled face toward the camera, and lets it go under a straight lift.

Workspace layout

Three rows: the pieces at the back, the count board and the bags in the middle, and the card, the pad and the finished bags at the front. Every piece travels forward out of the bin onto the board and then across into its bag, and every bag travels forward out of the rack onto the pad and then along into the slot with the same number.

Back row:

1. Bin spot, back centre on the midline, the bulk bin standing with its loose pieces spread across its floor.

Middle row, left to right:

2. Count board spot, left of the midline, the board lying flat with five squares in one line numbered 1 to 5 left to right, bare at the start of the episode.
3. Bag rack spot, right of the midline, four bags standing open in their rings in one line, numbered 1 to 4 left to right.
4. Label strip spot, at the right end of the middle row, the strip lying flat with four labels numbered 1 to 4, clear of the bag rack.

Front row, left to right:

5. Marker rest, at the left end of the front row, the marker lying flat with its tip toward the front of the table.
6. Tally card spot, left of the midline, the card lying flat with four numbered rows facing the camera.

7. Seal pad, right of the midline, straight forward of the bag rack, bare at the start of the episode.
8. Done rack, at the right end of the front row, four slots in one line numbered 1 to 4 left to right, empty at the start of the episode.
9. Spill strip, the bare table between the front row and the front table edge. It stays bare unless a piece falls onto it, and a piece that lands there is left where it lies.

A clear gap of at least a gripper's width sits between the bin and the board, between the board and the rack, between one bag and the next, and between one done rack slot and the next, so a piece lifts straight up and a bag lifts straight out without the gripper touching anything beside it. Each bag mouth is reached from the front of the rack, never across the top of another bag. No carry passes over an open bag, over a bag already in the done rack, over the tally card or over the label strip.

Vocabulary

- **Arm assignment:** one gripper does everything: every piece, every bag, every seal, every label and every stroke. This is a single arm task and no second arm takes any part in it.
- **Hold:** a closed empty gripper pressed onto an object to keep it still. Nothing in this task is held: every fixture grips the table and every bag stands open on its own.
- **Tool home:** the spot the marker returns to and lies flat on before the next step starts. The marker is never left on the tally card, on the seal pad, on the tabletop or in the gripper between steps.
- **Release point:** the gripper opens over a numbered square, over the mouth of the bag being filled, over the seal pad, over a done rack slot or over the marker rest. Never over the bin, never over another bag and never over open tabletop.
- **Stable placement:** the object stands still after release, flat or square on its surface, and does not rock, tip, slide or lean on anything.
- **Piece:** one of the small parts counted out of the bulk bin. Every piece is the same part, so one piece is as good as any other and no piece is chosen over another.
- **One piece per pick:** the gripper comes up out of the bin holding one piece and nothing else. Two pieces in the gripper is not a pick, it is a miscount waiting to happen.
- **Bulk bin:** the open bin the pieces are taken from. It is never lifted, never tipped and never emptied out.
- **Count board:** the flat board with five numbered squares that the count is laid out on. It is the count: the pieces are counted where they lie, in the open, before any of them goes into a bag.
- **Square:** one of the five numbered places on the count board. It holds one piece and one piece only.

- **Full board:** five pieces on the board, one lying in each of the five squares, with no square empty, no square holding two, and no piece lying between the squares or on the board edge.
- **Bare board:** no piece anywhere on the board surface. The board is bare before a count starts and bare again before a bag is sealed.
- **Bag:** one of the four poly bags that are filled. It hangs open in its own numbered ring in the bag rack until it is filled.
- **Seal strip:** the press to close strip running across a bag's mouth. It is open while the bag is being filled and closed for its whole length once the bag is sealed.
- **Seal pad:** the firm flat pad a bag is laid on to be sealed and labelled. It gives the press something to push against. Nothing else is ever set down on it.
- **Sealed bag:** the seal strip pressed shut along its whole length, with no open gap at either end, no piece caught in the strip and nothing able to fall out when the bag is stood up.
- **Label strip:** the backing sheet carrying the four numbered labels. Labels are peeled off it in order, 1 first and 4 last.
- **Labelled bag:** one label pressed flat on the bag's front face, below the seal strip, sticking down along its whole edge with no corner lifted and no fold in it.
- **Tally card:** the card with one numbered row per bag. Each row prints its bag number and the count 5 and ends in one empty box.
- **Row check:** reading the row with the same number as the bag on the pad and confirming the label number matches the row number and the row's count matches the five squares that were filled. The check comes before the stroke, never after it.
- **Stroke:** one straight mark drawn down inside a row's box. One stroke per row, drawn once the row check has passed.
- **Marked row:** a row carrying one stroke inside its box, the ink inside the box lines and not across the printed number or into the row above or below.
- **Done rack:** the rack of four numbered slots the finished bags stand in. A bag goes into the slot with the same number as its label.
- **Dropped piece:** a piece that lands anywhere but a square, a bag or the bin. It is left where it falls and cleared in the reset, never on camera.
- **Quarter turn:** the most any object is tilted or turned in the gripper. Taking a piece, lowering it into a bag, laying a bag flat and standing it up all stay inside it. Nothing is ever turned over in the air.

Steps

Step 1: Count five pieces onto the count board

Goal: five pieces on the count board, one lying in each of the five numbered squares, with the board holding nothing else.

Bags are worked in rack order, bag 1 first and bag 4 last. The board is filled square 1 to square 5 in order, so a reviewer can count along the line and see where the count got to at any moment. The board is bare before the first piece is laid down.

1.1 Take one piece from the bin

- **Check:** the board is bare, with no piece left on it from the last bag.
- With the **gripper**, come down straight from above into the bin and close on one piece, square to it, clear of the bin wall and the bin floor.
- Lift it straight up out of the bin until it is clear of the rim, before any sideways move.
- One piece comes up and nothing else. If two pieces come up together, lower them back into the bin, open the gripper, and come down again for one.

Check: one piece, and one only, hangs in the gripper clear of the bin, and the bin has not moved. Make no more than two corrective takes on one piece. If the gripper still cannot come up with one piece, end the episode and correct the pieces during reset.

1.2 Lay it in the next empty square

- Carry the piece forward and left to the count board, keeping it level, and hold it above the next empty square in order: square 1 first, then 2, 3, 4 and 5.
- Lower it straight down until it lies on the square under its own weight, then open the gripper and lift straight clear.
- One piece to a square. Do not lay a second piece on a filled square, do not lay a piece between the squares or on the board edge, and do not slide a piece along the board into place.

Check: the piece lies still and fully inside its own square, with clear board showing round it, and no square holds two. If it has landed outside its square or across a line, lift it straight up clear of the board and lay it down again. Make no more than one correction per piece.

1.3 Fill all five squares

- Come back empty to the bin and take the next piece.
- Run 1.1 and 1.2 again until all five squares hold one piece each.
- Five pieces, taken one at a time, laid down one at a time, in square order. A piece is lying in its square before the next piece is picked up.

- If a piece falls off the board or off the table, leave it where it lands and take one fresh piece from the bin for that square. Make no more than one replacement piece per bag.

Check: the board is full, five pieces in five squares, one to a square, and the bin, the board and the bag rack have not moved. Count along the line before anything goes into the bag. If the board holds four or six, end the episode and let review flag it: the count is not corrected by adding or taking away once the move has started.

Step 2: Move the counted pieces into the bag

Goal: all five pieces inside the bag with the same number as this bag in the order, the board bare, and the bag still standing open in its ring.

The board is emptied square 1 to square 5 in order, one piece per carry, so the pieces come off the board in the same order they went onto it. The bag stands open on its own, so nothing holds it while the pieces go in.

2.1 Move one piece at a time, square 1 to square 5

- With the **gripper**, come down straight from above on the piece in square 1 and close on it square.
- Lift it straight up clear of the board, carry it across to the bag being filled, coming round the front of the rack rather than across the top of another bag.
- Bring it over the bag mouth, inside the ring of the mouth and low to it, and lower it inside the bag until it is below the mouth, then open the gripper and lift straight clear.
- The piece is let go inside the bag, not dropped from above the mouth. The gripper does not touch the bag walls and does not go below the seal strip more than it has to.
- Repeat for square 2, then 3, 4 and 5, one piece per carry, in square order.

Check: after each carry the square it came from is empty and the piece is in the bag. If a piece lands outside the bag, leave it where it falls and end the episode: a bag that has taken four pieces is not topped up from the bin.

2.2 Check the board and the bag

- Look at the board: all five squares empty and no piece anywhere on the board surface.
- Look into the bag: five pieces lying in the bottom of it, well below the seal strip, and nothing caught in the strip or standing up out of the mouth.
- Look at the rack: the other bags still standing open, empty and untouched.

Check: the board is bare and the bag holds five. If a piece is still on the board, move it now, before the bag leaves the rack. Once the bag is off the rack nothing else goes into it.

Step 3: Seal the bag

Goal: the bag lying flat on the seal pad with its seal strip closed along its whole length.

The bag is sealed lying down, not standing up. The pad gives the press something to push against, and a bag lying flat keeps its five pieces away from the strip.

3.1 Lift the bag onto the seal pad

- With the **gripper**, close on the bag just below its seal strip, square across the mouth, so the mouth is held together as it comes up.
- Lift the bag straight up out of its ring until it is clear of the rack, before any sideways move, and keep it upright so no piece runs out of the mouth.
- Carry it forward to the seal pad, lower it until the bottom of the bag rests on the pad, and lay it down flat with the seal strip along the top edge and the front face upward.
- Open the gripper and lift straight clear.

Check: the bag lies flat and still on the pad, fully on it, with the seal strip straight along the top edge, the front face upward and the five pieces spread in the bottom of the bag away from the strip. If the bag lies folded under itself or half off the pad, lift it clear and lay it down again. Make no more than one correction.

3.2 Press the seal strip closed

- Close the gripper flat and bring it down square on one end of the seal strip, pressing the two walls of the strip together against the pad.
- Draw the closed gripper along the strip to the other end in one steady move, in contact the whole way, so the strip closes behind it as it goes.
- One pass, end to end. Do not press one part and then jump to another, do not go back and forth along the strip, and do not press with the gripper open.
- Lift the gripper straight off the strip at the end of the pass.

Check: look along the strip. It is closed for its whole length, with no open gap at either end and no piece caught in it. If a short part of the strip is still open, make one more pass over that part only. If the strip will not close, end the episode and correct the bag during reset. A sealed bag is never opened again on camera.

Step 4: Label the bag

Goal: the label with the same number as this bag pressed flat on the bag's front face.

Labels are peeled in order, 1 first and 4 last, so the label number always matches the bag number in the rack order. The bag stays lying on the pad while the label goes on.

4.1 Take the next label off the strip

- With the **gripper**, come down on the edge of the next label in order and close on that edge only, clear of the backing beside it.
- Lift straight up to peel the label off its backing, keeping it flat and sticky side down, and do not let it curl onto itself.

- Carry it to the bag on the pad, keeping it level and sticky side down the whole way.

Check: one label hangs flat in the gripper, sticky side down, its number readable, and the backing lies flat on its spot with the labels after it untouched. If the label has folded onto itself, end the episode: a folded label is not unfolded and used.

4.2 Press it on the bag's front face

- Hold the label flat above the bag's front face, square to the bag, below the seal strip and clear of the pieces inside.
- Lower it straight down until it touches the bag, then open the gripper and lift clear.
- Close the gripper flat and press straight down once across the label, so it sticks down along its whole edge.
- Press straight down and nothing else. Do not rub the label, do not slide it into place once it has touched the bag, and do not peel it back off to line it up again.

Check: the label lies flat on the front face, square to the bag, below the seal strip, with no corner lifted, no fold in it and its number readable from the camera. If a corner is lifted, press it down once more with the flat of the gripper. Make no more than one correction.

Step 5: Verify the tally card and mark it

Goal: the row with the same number as this bag read and confirmed, and one stroke drawn in its box, with the marker back on its rest.

The card is the record of the count, so the check comes before the stroke. A stroke says five pieces went into the bag with this number: it is drawn once the row has been read against the bag, and never before.

5.1 Read the row for this bag

- Look at the tally card and find the row with the same number as the label on the bag: bag 1 to row 1, bag 4 to row 4.
- Confirm the label number on the bag and the row number are the same, that the row's count reads 5, and that five squares were filled on the board for this bag.
- Confirm the row's box is still empty and no other row has been marked out of order.

Check: the row matches the bag. If the label number and the row number do not match, or the row is already marked, end the episode and let review flag it. No stroke is drawn and no label is changed.

5.2 Draw one stroke in the row's box

- With the **gripper**, close on the marker on its rest, square across the barrel, and lift it straight up off the rest with the tip pointing down.
- Carry it to the tally card and hold the tip above the empty box in that bag's row, clear of the printed number.

- Lower the tip until it touches the card and draw one straight stroke down inside the box, then lift the tip straight off the card.
- One stroke, inside the box. Do not go over the stroke again, do not draw across the printed number, and do not mark a row for a bag that has not been sealed and labelled.

Check: the row carries one stroke, the ink is inside the box lines, and no other row has been marked. If the stroke has gone outside the box, leave it and let review flag it. A stroke is never rubbed out, drawn over or crossed through on camera.

5.3 Park the marker

- Carry the marker back to its rest, lower it until it lies flat with the tip toward the front of the table, release, and lift clear.

Check: the marker lies fully on its rest, not on the tally card, on the seal pad or on the tabletop, and the card has not moved.

Step 6: Stand the bag in the done rack

Goal: the sealed, labelled bag standing in the done rack slot with the same number as its label, and the seal pad bare again.

6.1 Lift the bag off the pad and stand it in its slot

- With the **gripper**, close on the bag just below its seal strip, square across the sealed mouth, and lift it straight up off the pad.
- Carry it along the front row to the done rack slot with the same number as its label, keeping it clear of the bags already standing there.
- Lower it into its slot until it stands on its own with the labelled face toward the camera, then open the gripper and lift clear.
- One bag per slot. The bag is not pushed into the slot sideways and it is not leaned against the bag beside it.

Check: the bag stands upright and still in the slot with its own number, its label readable from the camera, the seal still closed, and the seal pad bare. If the bag leans, rocks or sits in the wrong slot, lift it straight up clear and stand it again. Make no more than one correction per bag.

Step 7: Do the next bag

- Come back empty to the bin and start the count for the next bag in rack order.
- Run Steps 1 to 6 again for that bag.
- Four bags, worked one at a time from bag 1 to bag 4. A bag is sealed, labelled, marked and racked before the next bag is touched.
- The numbers stay in step: bag 1 takes label 1, row 1 and slot 1, and bag 4 takes label 4, row 4 and slot 4.
- The board is bare again before the next count starts.

Check: after each bag, the rack holds one fewer open bag, the done rack holds one more sealed bag, the label strip has one fewer label, the tally card carries one more stroke, and the board is bare. No bag is filled twice or skipped.

Step 8: End the episode

- Confirm the board: bare, with no piece left on it or between the squares.
- Confirm the bag rack: all four rings empty.
- Confirm the done rack: four bags, one standing in each numbered slot, each one sealed along the whole strip and each label readable and matching its slot number.
- Confirm the label strip: all four labels peeled, the backing lying flat on its spot.
- Confirm the tally card: one stroke in every row, four strokes in all, each inside its own box.
- Confirm the tools: the marker flat on its rest and the seal pad bare.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return the arm home with its gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the four bags out of the done rack by hand, open each seal strip, tip the pieces back into the bulk bin, and count them as they go in: every bag must give back five. Set aside any bag that gave back a different number and note it for review.
2. Peel the four labels off the bags and bin them, then bin the used bags. A bag that has been sealed once is not filled again.
3. Hang four fresh poly bags in the rack rings, one to a ring, and check each one stands open on its own with its seal strip open and clear above the ring.
4. Pick up any piece that fell onto the spill strip, onto the tabletop or onto the floor, wipe it, and drop it back into the bin. A piece that has been on the floor and cannot be wiped clean is binned.
5. Top up the bin so it holds well more than twenty pieces, spread across its floor rather than heaped in one corner, and pick out any piece that is bent, broken or stuck to another.
6. Lay a fresh label strip flat on its spot with four labels numbered 1 to 4 and the numbers facing the camera, and peel one spare label by hand to confirm the strip still peels clean.
7. Lay a fresh blank tally card on its spot with four numbered rows facing the camera and every box empty.
8. Wipe the count board, the seal pad and the tabletop clean and dry, and confirm the board is bare and the pad is bare.

9. Lay the marker flat on its rest with the cap off and the tip toward the front of the table.
Replace a marker that skips or will not mark poly under one stroke.
10. Confirm the bin, the board, the bag rack, the seal pad, the tally card holder and the done rack all still grip the table, and that the square numbers, the bag numbers, the label numbers, the row numbers and the slot numbers all still face the camera.
11. Confirm the done rack is empty and every slot still holds a bag upright on its own.
12. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a piece goes into a bag before all five squares are full, a bag is lifted off the rack before the board is bare, a bag is labelled before it is sealed, a stroke is drawn before the bag is sealed and labelled, a bag is stood in the done rack before its row is marked, a new count is started while the last bag is still on the seal pad, or a step is skipped.
- **SOP rule broken:** Steps 1 to 6, for each bag count five onto the board, move them in, seal it, label it, verify the row and mark it, then rack it, in that order.
- **Coaching note:** count it, fill it, seal it, label it, mark it, rack it. One bag all the way through before the next one is touched.

Violation: A second arm used

- **Visible cue:** a second arm enters the work area, closes its gripper, touches an object, steadies the bin, the board, a bag, the card or the rack, or presses a seal or a label. Any object taken, carried or pressed by a second gripper is a violation on its own.
- **SOP rule broken:** Steps 1 to 8, this is a single arm task: one gripper does all of it and nothing else in the cell touches the task.

- **Coaching note:** one arm only. If a second arm has worked, the episode is a bimanual episode wearing the wrong label.

Violation: Object held or steadied

- **Visible cue:** the gripper presses on the bin, the board, the bag rack, the seal pad, the tally card or the done rack to keep it still while working, or holds a bag mouth open while pieces go in, instead of letting the table and the rings do it.
- **SOP rule broken:** Steps 1 to 6, nothing in this task is held: every fixture grips the table and every bag stands open on its own in its ring.
- **Coaching note:** if something has to be held down, it was not set up right. Let go and let the fixture do its job.

Violation: More than one piece picked at once

- **Visible cue:** the gripper comes up out of the bin holding two or more pieces, two pieces are carried in one trip, or two pieces are let go into the bag in one move.
- **SOP rule broken:** Step 1.1, one piece comes out of the bin per pick, and Step 2.1, one piece per carry into the bag.
- **Coaching note:** one piece in the gripper, every time. Two pieces at once is where a miscount starts, and nobody can count it back on camera.

Violation: Count laid out wrong on the board

- **Visible cue:** a square is skipped or filled out of order, two pieces are laid on one square, a piece is laid between the squares or on the board edge, a piece is slid along the board into place, or the move into the bag starts with a square still empty.
- **SOP rule broken:** Step 1.2 and Step 1.3, one piece to a square, square 1 to square 5 in order, and all five squares full before anything goes into the bag.
- **Coaching note:** the board is the count. A piece off its square is a piece nobody can count.

Violation: Wrong number of pieces in the bag

- **Visible cue:** fewer or more than five pieces go into a bag, a sixth piece is added to top up a short count, a piece is taken back out of a bag, or a piece is put into a bag straight from the bin without being laid on the board first.
- **SOP rule broken:** Steps 1 and 2, five pieces per bag, counted on the board first and moved in one at a time from square 1 to square 5.
- **Coaching note:** the count is fixed at five and it is laid out in the open before it goes in. A correction piece makes the episode unlearnable.

Violation: Piece dropped or put outside the bag

- **Visible cue:** a piece is let go above the bag mouth instead of inside the bag, a piece lands on the rack, on the tabletop or on the spill strip, or a piece falls out of a bag as it is lifted.

- **SOP rule broken:** Step 2.1, the piece is lowered inside the bag below the mouth before the gripper opens, and Step 3.1, the bag is kept upright as it comes out of its ring.
- **Coaching note:** carry it all the way in, then let go. A piece dropped from above the mouth is as likely to land outside as in.

Violation: Bag filled or lifted wrong

- **Visible cue:** the wrong bag in the rack order is filled, a piece is carried across the top of another bag, the gripper touches or knocks an open bag beside the one being filled, a bag is lifted off the rack before it holds five, or a bag is tipped, swung or carried on its side with its mouth still open.
- **SOP rule broken:** Steps 2.1 and 3.1, bags are filled in rack order, each mouth reached from the front of the rack, and a bag leaves its ring upright and only once it holds five.
- **Coaching note:** round the front of the rack, never over the top. The mouth stays up until the strip is shut.

Violation: Seal wrong, or sealed more than once

- **Visible cue:** the strip is pressed while the bag is standing up or held in the air instead of lying on the pad, the gripper is drawn back and forth along the strip, the press starts part way along and leaves an end open, a piece is caught in the strip, or a sealed bag is opened again on camera.
- **SOP rule broken:** Step 3.2, one steady pass with the closed gripper from one end of the strip to the other, pressed against the pad, and a sealed bag is never opened again.
- **Coaching note:** lay it flat, press one end, run to the other. Look along the strip before the bag leaves the pad.

Violation: Label wrong, or put on wrong

- **Visible cue:** a label is peeled out of order, two labels are peeled at once, a label goes on a bag whose number it does not match, a label is put on over the seal strip or on the back of the bag, a label is rubbed or slid into place after it has touched the bag, or a label is peeled back off and put on again.
- **SOP rule broken:** Step 4, labels are peeled in order 1 to 4, one at a time, and each one is lowered square onto the bag's front face below the seal strip and pressed straight down once.
- **Coaching note:** peel in order, land it once, press it flat. A label that has to be moved was not lined up before it touched.

Violation: Tally card marked wrong

- **Visible cue:** a stroke is drawn in the wrong row, more than one stroke goes in a row, a stroke goes outside its box or across the printed number, a row is marked for a bag that is

not sealed and labelled, all the rows are marked at the end in one go, or a stroke is rubbed out, drawn over or crossed through.

- **SOP rule broken:** Step 5, read the row against the bag first, then draw one stroke inside that row's box, one row per bag as each bag is finished.
- **Coaching note:** read it, then mark it, one row at a time. The card is the record, so nothing on it is ever undone.

Violation: Verify skipped or done after the fact

- **Visible cue:** the stroke goes down without the label number and the row number being brought together, a row is marked while the bag is still in the rack or already in the done rack, or a mismatch between the label and the row is marked anyway.
- **SOP rule broken:** Step 5.1, the label number, the row number and the count of five are confirmed while the bag lies on the pad, and only then is the stroke drawn.
- **Coaching note:** the check is the point of the card. A stroke drawn without it records nothing.

Violation: Bag racked wrong

- **Visible cue:** a bag goes into a slot whose number is not its label number, a bag is pushed into a slot sideways or leaned against the bag beside it, two bags go in one slot, a bag stands with its label away from the camera, or a bag already in the done rack is lifted out and handled again.
- **SOP rule broken:** Step 6, each bag stands on its own in the slot with the same number as its label, labelled face toward the camera, and racked bags are not touched again.
- **Coaching note:** label number and slot number are the same number, every time. Once it is standing, leave it.

Violation: Spill cleaned up

- **Visible cue:** after a piece lands outside a bag, on the board edge, on the spill strip or on the floor, the gripper is used to sweep, push, brush or pick it up, or a piece is picked up off the floor.
- **SOP rule broken:** Steps 1 to 6, a dropped piece is left where it falls and is cleared in the reset.
- **Coaching note:** cleanup motions are not part of the task. The spill is data, the cleanup is noise.

Violation: A carry passed over something it should not

- **Visible cue:** a piece passes over an open bag it is not going into, a bag passes over the tally card, the marker or a bag already in the done rack, or the marker passes over the seal pad or an open bag.

- **SOP rule broken:** Steps 1 to 6, each piece goes round the front of the rack to its own bag, and nothing is carried over an open bag, over a racked bag or over the tally card.
- **Coaching note:** route around, not across. A piece dropped into the wrong bag is two miscounts at once.

Violation: Bad pick, carry, or release

- **Visible cue:** an object is picked up before the last one is set down, two objects are held at once, the gripper shifts or seats its hold again while the object is off the table, a piece or a bag or the marker slides along the table with no daylight under it, a bag hangs tilted because the grip was taken off square, or an object is turned past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 6, one object at a time, lifted straight up clear of the surface before any horizontal motion, gripped square, and tilted no more than a quarter turn.
- **Coaching note:** set it down to change the grip, and lift before the carry. A bag that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** a piece, a bag, the board, the card, the label strip or the marker is shoved, nudged or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 6, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Tool not returned to its home

- **Visible cue:** the gripper opens while the marker is still above its rest, the marker is left on the tally card, on the seal pad or on the tabletop, or it is still in the gripper when the next step begins.
- **SOP rule broken:** Step 5.3, the marker returns to its rest and lies flat there before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower it until it lies flat, then release.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, the gripper or a carried object strikes the bin, the board, the bag rack, an open bag, the seal pad, the tally card or a bag already in the done rack and knocks, tips, slides or displaces it; a bag falls off the pad or out of the gripper; or a piece lands short of its square, with no gripper failure, drift, or controller fault visible.

- **SOP rule broken:** Steps 1 to 6, keep each object on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. The open bags are the thing worth steering widest of.

Violation: Manipulation outside the camera frame

- **Visible cue:** a pick, a lay down, a move into a bag, a seal, a label, a stroke or a racking happens partly or wholly outside the environment camera frame, or the arm stands between the camera and the board while the count is what has to be seen.
- **SOP rule broken:** Steps 1 to 6, every manipulation happens inside the camera frame, and the board, the bag being filled and the tally card stay visible.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions and keep the arm clear of the board.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a piece still on the board, a bag still open in the rack, a bag unsealed or unlabelled, a bag in the wrong slot, a row unmarked or marked twice, a label left on the strip, the marker off its rest, the arm short of home, or the gripper not fully open.
- **SOP rule broken:** Step 8, confirm the board, the rack, the done rack, the label strip, the card and the tools, then return the arm home with its gripper open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arm does before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside the arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective pieces at episode start: pieces that stick, nest or hook together so two come up as one, or pieces too small to be seen as one piece on camera. Replace the batch.
- Defective count board: one that slides, turns or lifts under a correct pick, one whose square numbers cannot be read, or one whose squares are wide enough to hold two pieces side by side.

- Defective bag: one that will not stand open in its ring, one with a tear or a hole, or one whose seal strip will not close along its whole length under a correct pass.
- Defective bag rack: one that grips a bag so hard it cannot be lifted straight out, or one whose rings let a bag close as it is filled.
- Defective seal pad: one that slides under a correct press, or that is too soft to press a strip against.
- Defective label or label strip: a label that tears, that lifts its backing with it, that will not stick to poly under one press, or a strip whose numbers cannot be read.
- Defective marker: one that skips, that will not mark poly or card under one stroke, or that has dried out.
- Defective done rack: one that will not hold a sealed bag upright on its own, or that grips a bag so it cannot be stood in the slot under a straight lower.
- Glare, shadow or a light flickering over the count board or the seal pad that stops the pieces, the seal or the ink being seen. Fix the lighting before the next episode.
- Bin holding fewer than twenty pieces at episode start, a rack holding fewer or more than four bags, a strip holding fewer than four labels, a bag that already has something in it, or a tally card with a box already marked. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take one piece from the bulk bin
2. Lay one piece in a numbered square
3. Take one piece off the count board
4. Put one piece into the bag
5. Lift the bag out of the bag rack
6. Lay the bag flat on the seal pad
7. Press the seal strip closed
8. Peel one label off the strip
9. Press one label onto the bag
10. Take the marker from its rest
11. Draw one stroke in a tally card row
12. Return the marker to its rest
13. Stand one bag in its done rack slot
14. Return the arm home and end the episode